Displaying data from multiple tables using joins

1) Write a query for the HR department to produce the addresses of all the departments. Use the EMP and DEPT tables. Show the EMPNO, ENAME, SAL, DNAME and LOC in the output. Use a NATURAL JOIN to produce the results.

```
SELECT EMPNO, ENAME, SAL, DNAME, LOC
FROM EMP
NATURAL JOIN DEPT;
```

Output:-

```
QL> SELECT EMPNO, ENAME, SAL, DNAME, LOC
 2  FROM EMP
 3  NATURAL JOIN DEPT;

   EMPNO ENAME                       SAL DNAME
---------- -------------------- ---------- --------------------
OC
-------------------
     102 SCOTT                      3000 IT
ANGALORE

     103 TURNER                     1500 SALES
HENNAI

     104 SMITH                       800 IT
ANGALORE

   EMPNO ENAME                       SAL DNAME
---------- -------------------- ---------- --------------------
OC
-------------------
     105 BLAKE                      2850 SALES
HENNAI

     106 JONES                      2975 IT
ANGALORE

     107 ALLEN                      1600 SALES
HENNAI

   EMPNO ENAME                       SAL DNAME
---------- -------------------- ---------- --------------------
OC
-------------------
     108 WARD                       1250 SALES
HENNAI

     109 MARTIN                     1250 SALES
HENNAI

     110 JAMES                       950 SALES
HENNAI

   EMPNO ENAME                       SAL DNAME
---------- -------------------- ---------- --------------------
OC
-------------------
     111 MILLER                     1300 HR
YDERABAD

     112 CLARK                      2450 HR
YDERABAD

     113 ADAMS                      1100 IT
ANGALORE

   EMPNO ENAME                       SAL DNAME
---------- -------------------- ---------- --------------------
OC
-------------------
     114 SARAH                      2100 HR
YDERABAD

     115 RAMESH                     2600 IT
ANGALORE

     101 KING                       5000 HR
YDERABAD

5 rows selected.
```

2) The HR department needs a report of all employees. Write a query to display the JOB, MGR, SAL, COMM, DNAME of employees whose JOB is SALESMAN.

```
SELECT JOB, MGR, SAL, COMM, DNAME
FROM EMP, DEPT
WHERE EMP.DEPTNO = DEPT.DEPTNO
AND JOB = 'SALESMAN';
```

Output:-

```
SQL> SELECT JOB, MGR, SAL, COMM, DNAME
  2  FROM EMP, DEPT
  3  WHERE EMP.DEPTNO = DEPT.DEPTNO
  4  AND JOB = 'SALESMAN';

JOB                         MGR        SAL       COMM DNAME
-------------------- ---------- ---------- ---------- --------------------
SALESMAN                    105       1500          0 SALES
SALESMAN                    105       1600        300 SALES
SALESMAN                    105       1250        500 SALES
SALESMAN                    105       1250       1400 SALES

SQL>
```

3) The HR department needs a report of employees in LOC DALLAS. Display the ENAME, JOB, DEPTNO, and DNAME for all employees who work in DALLAS.

```
SELECT ENAME, JOB, EMP.DEPTNO, DNAME
FROM EMP, DEPT
WHERE EMP.DEPTNO = DEPT.DEPTNO
AND LOC = 'DALLAS';
```

Output:-

```
SQL> SELECT ENAME, JOB, EMP.DEPTNO, DNAME
  2  FROM EMP, DEPT
  3  WHERE EMP.DEPTNO = DEPT.DEPTNO
  4  AND LOC = 'DALLAS';

no rows selected

SQL>
```

4) Create a report to display employees' name and employee number along with their manager's name and manager number. Label the columns Employee, Emp#, Manager, and Mgr#, respectively.

```
SELECT E.ENAME as Employee,
    E.EMPNO as Emp#,
    M.ENAME as Manager,
    M.EMPNO as Mgr#
FROM EMP E, EMP M
WHERE E.MGR = M.EMPNO;
```

Output:-

```
SQL> SELECT E.ENAME as Employee,
  2         E.EMPNO as Emp#,
  3         M.ENAME as Manager,
  4         M.EMPNO as Mgr#
  5  FROM EMP E, EMP M
  6  WHERE E.MGR = M.EMPNO;

EMPLOYEE                   EMP# MANAGER                    MGR#
-------------------- ---------- -------------------- ----------
ADAMS                       113 SCOTT                       102
TURNER                      103 BLAKE                       105
ALLEN                       107 BLAKE                       105
WARD                        108 BLAKE                       105
MARTIN                      109 BLAKE                       105
JAMES                       110 BLAKE                       105
SCOTT                       102 JONES                       106
SMITH                       104 JONES                       106
SARAH                       114 CLARK                       112
RAMESH                      115 CLARK                       112
BLAKE                       105 KING                        101

EMPLOYEE                   EMP# MANAGER                    MGR#
-------------------- ---------- -------------------- ----------
JONES                       106 KING                        101
MILLER                      111 KING                        101
CLARK                       112 KING                        101

14 rows selected.

SQL>
```

5) Modify the previous query to display all employees including King, who has no manager. Order the results by the employee number.

```
SELECT E.ENAME as Employee,
    E.EMPNO as Emp#,
    M.ENAME as Manager,
    M.EMPNO as Mgr#
FROM EMP E, EMP M
WHERE E.MGR = M.EMPNO(+)
ORDER BY E.EMPNO;
```

Output:-

```
SQL> SELECT E.ENAME as Employee,
  2         E.EMPNO as Emp#,
  3         M.ENAME as Manager,
  4         M.EMPNO as Mgr#
  5  FROM EMP E, EMP M
  6  WHERE E.MGR = M.EMPNO(+)
  7  ORDER BY E.EMPNO;

EMPLOYEE                   EMP# MANAGER                    MGR#
-------------------- ---------- -------------------- ----------
KING                        101
SCOTT                       102 JONES                       106
TURNER                      103 BLAKE                       105
SMITH                       104 JONES                       106
BLAKE                       105 KING                        101
JONES                       106 KING                        101
ALLEN                       107 BLAKE                       105
WARD                        108 BLAKE                       105
MARTIN                      109 BLAKE                       105
JAMES                       110 BLAKE                       105
MILLER                      111 KING                        101

EMPLOYEE                   EMP# MANAGER                    MGR#
-------------------- ---------- -------------------- ----------
CLARK                       112 KING                        101
ADAMS                       113 SCOTT                       102
SARAH                       114 CLARK                       112
RAMESH                      115 CLARK                       112

15 rows selected.

SQL> |
```

6) The HR department needs a report on job grades and salaries. To familiarize yourself with the SALGRADE table, first show the structure of the SALGRADE table. Then create a query that displays the name, job, department name, salary, and grade for all employees.

Describe SALGRADE;

Output:-

```
SQL> DESC SALGRADE;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 GRADE                                              NUMBER
 LOSAL                                              NUMBER
 HISAL                                              NUMBER

SQL> SELECT ENAME, JOB, DNAME, SAL, GRADE
```

```
SELECT ENAME, JOB, DNAME, SAL, GRADE
FROM EMP, DEPT, SALGRADE
WHERE EMP.DEPTNO = DEPT.DEPTNO
AND SAL BETWEEN LOSAL AND HISAL;
```

Output:-

```
SQL> SELECT ENAME, JOB, DNAME, SAL, GRADE
  2  FROM EMP, DEPT, SALGRADE
  3  WHERE EMP.DEPTNO = DEPT.DEPTNO
  4  AND SAL BETWEEN LOSAL AND HISAL;

ENAME                JOB                  DNAME                       SAL
-------------------- -------------------- -------------------- ----------
     GRADE
----------
SCOTT                ANALYST              IT                         3000
         4

TURNER               SALESMAN             SALES                      1500
         3

SMITH                CLERK                IT                          800
         1


ENAME                JOB                  DNAME                       SAL
-------------------- -------------------- -------------------- ----------
     GRADE
----------
BLAKE                MANAGER              SALES                      2850
         4

JONES                MANAGER              IT                         2975
         4

ALLEN                SALESMAN             SALES                      1600
         3


ENAME                JOB                  DNAME                       SAL
-------------------- -------------------- -------------------- ----------
     GRADE
----------
WARD                 SALESMAN             SALES                      1250
         2

MARTIN               SALESMAN             SALES                      1250
         2

JAMES                CLERK                SALES                       950
         1


ENAME                JOB                  DNAME                       SAL
-------------------- -------------------- -------------------- ----------
     GRADE
----------
MILLER               CLERK                HR                         1300
         2

CLARK                MANAGER              HR                         2450
         4

ADAMS                CLERK                IT                         1100
         1


ENAME                JOB                  DNAME                       SAL
-------------------- -------------------- -------------------- ----------
     GRADE
----------
SARAH                CLERK                HR                         2100
         4

RAMESH               ANALYST              IT                         2600
         4

KING                 PRESIDENT            HR                         5000
         5


15 rows selected.
```

7) Display the ENAME, DNAME of all the employees. Also display those department name which do not have any employees working.

```
SELECT ENAME, DNAME
FROM EMP, DEPT
WHERE EMP.DEPTNO(+) = DEPT.DEPTNO;
```

<u>Output:-</u>

```
SQL> SELECT ENAME, DNAME
  2  FROM EMP, DEPT
  3  WHERE EMP.DEPTNO(+) = DEPT.DEPTNO;

ENAME                DNAME
-------------------- --------------------
SCOTT                IT
TURNER               SALES
SMITH                IT
BLAKE                SALES
JONES                IT
ALLEN                SALES
WARD                 SALES
MARTIN               SALES
JAMES                SALES
MILLER               HR
CLARK                HR

ENAME                DNAME
-------------------- --------------------
ADAMS                IT
SARAH                HR
RAMESH               IT
KING                 HR

15 rows selected.

SQL> |
```

8) The HR department needs to find the names and hire dates for all employees who were hired before their managers, along with their managers' names and hire dates.

```
SELECT E.ENAME,
    E.HIREDATE,
    M.ENAME,
    M.HIREDATE
FROM EMP E, EMP M
WHERE E.MGR = M.EMPNO
AND E.HIREDATE < M.HIREDATE;
```

<u>Output:-</u>

```
SQL> SELECT E.ENAME,
  2         E.HIREDATE,
  3         M.ENAME,
  4         M.HIREDATE
  5  FROM EMP E, EMP M
  6  WHERE E.MGR = M.EMPNO
  7  AND E.HIREDATE < M.HIREDATE;

ENAME                HIREDATE  ENAME                HIREDATE
-------------------- --------- -------------------- ---------
ALLEN                20-FEB-81 BLAKE                01-MAY-81
WARD                 22-FEB-81 BLAKE                01-MAY-81
SMITH                17-DEC-80 JONES                02-APR-81
BLAKE                01-MAY-81 KING                 17-NOV-81
JONES                02-APR-81 KING                 17-NOV-81
CLARK                09-JUN-81 KING                 17-NOV-81

6 rows selected.

SQL> |
```

9) Display the EMPNO, ENAME, DNAME, LOC of those employees who are working as CLERK. Use the USING clause.

```
SELECT EMPNO, ENAME, DNAME, LOC
FROM EMP
JOIN DEPT
USING (DEPTNO)
WHERE JOB = 'CLERK';
```

Output:-

```
SQL> SELECT EMPNO, ENAME, DNAME, LOC
  2  FROM EMP
  3  JOIN DEPT
  4  USING (DEPTNO)
  5  WHERE JOB = 'CLERK';

     EMPNO ENAME                DNAME                LOC
---------- -------------------- -------------------- --------------------
       104 SMITH                IT                   BANGALORE
       110 JAMES                SALES                CHENNAI
       111 MILLER               HR                   HYDERABAD
       113 ADAMS                IT                   BANGALORE
       114 SARAH                HR                   HYDERABAD

SQL>
```

10) Display the ENAME, SAL, MGR, DNAME of employees whose salary is more than 2000. Use the ON clause.

```
SELECT ENAME, SAL, MGR, DNAME
FROM EMP
JOIN DEPT
ON EMP.DEPTNO = DEPT.DEPTNO
WHERE SAL > 2000;
```

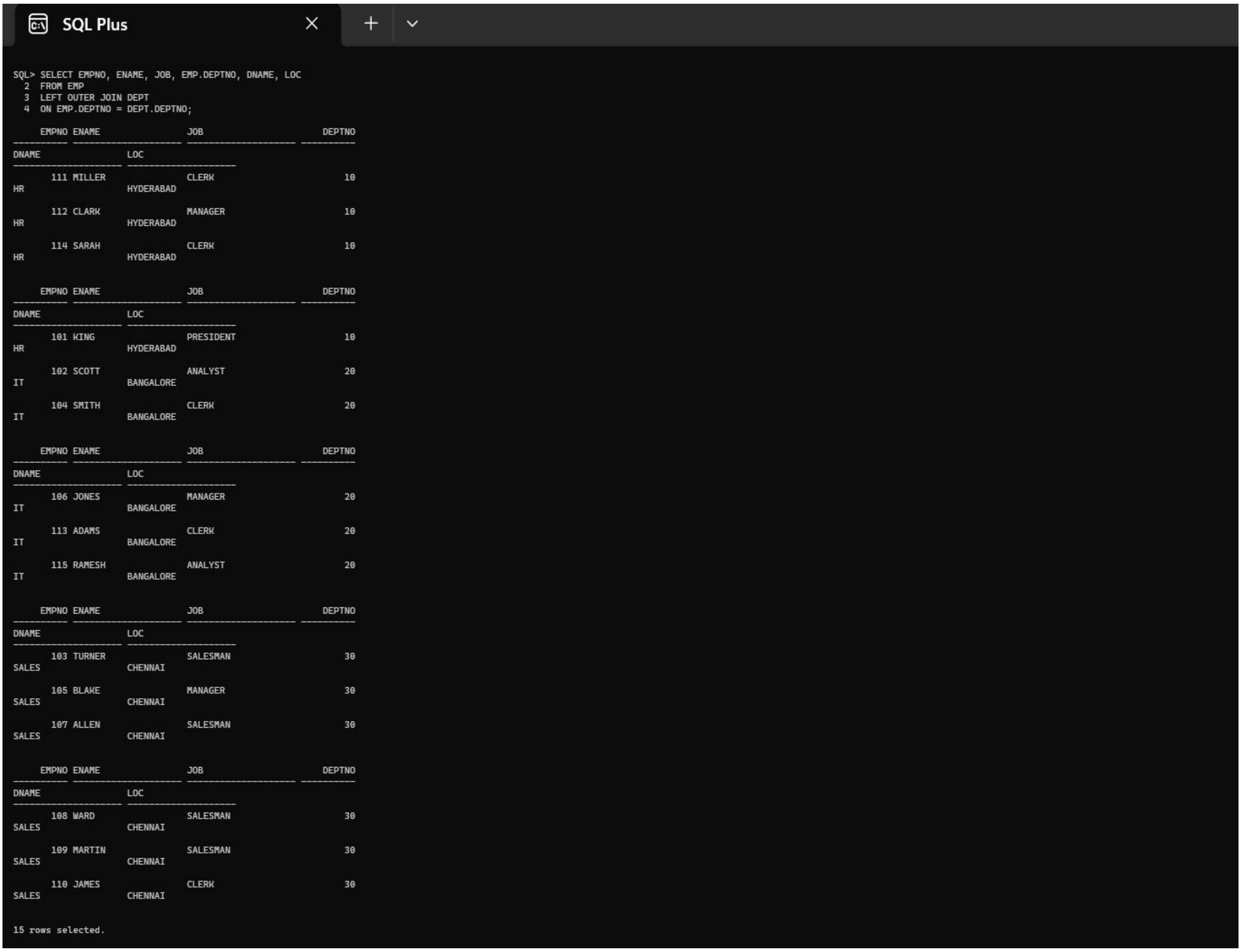
Output:-

```
SQL> SELECT ENAME, SAL, MGR, DNAME
  2  FROM EMP
  3  JOIN DEPT
  4  ON EMP.DEPTNO = DEPT.DEPTNO
  5  WHERE SAL > 2000;

ENAME                       SAL        MGR DNAME
-------------------- ---------- ---------- --------------------
SCOTT                      3000        106 IT
BLAKE                      2850        101 SALES
JONES                      2975        101 IT
CLARK                      2450        101 HR
SARAH                      2100        112 HR
RAMESH                     2600        112 IT
KING                       5000            HR

7 rows selected.

SQL> |
```

11) Display the EMPNO, ENAME, JOB, DEPTNO, DNAME, LOC of employees. Use LEFT OUTER JOIN.

```
SELECT EMPNO, ENAME, JOB, EMP.DEPTNO, DNAME, LOC
FROM EMP
LEFT OUTER JOIN DEPT
ON EMP.DEPTNO = DEPT.DEPTNO;
```

<u>Output:-</u>



```
SQL Plus

SQL> SELECT EMPNO, ENAME, JOB, EMP.DEPTNO, DNAME, LOC
  2  FROM EMP
  3  LEFT OUTER JOIN DEPT
  4  ON EMP.DEPTNO = DEPT.DEPTNO;

     EMPNO ENAME                JOB                      DEPTNO
---------- -------------------- -------------------- ----------
DNAME                LOC
-------------------- --------------------
       111 MILLER               CLERK                        10
HR                   HYDERABAD

       112 CLARK                MANAGER                      10
HR                   HYDERABAD

       114 SARAH                CLERK                        10
HR                   HYDERABAD


     EMPNO ENAME                JOB                      DEPTNO
---------- -------------------- -------------------- ----------
DNAME                LOC
-------------------- --------------------
       101 KING                 PRESIDENT                    10
HR                   HYDERABAD

       102 SCOTT                ANALYST                      20
IT                   BANGALORE

       104 SMITH                CLERK                        20
IT                   BANGALORE


     EMPNO ENAME                JOB                      DEPTNO
---------- -------------------- -------------------- ----------
DNAME                LOC
-------------------- --------------------
       106 JONES                MANAGER                      20
IT                   BANGALORE

       113 ADAMS                CLERK                        20
IT                   BANGALORE

       115 RAMESH               ANALYST                      20
IT                   BANGALORE


     EMPNO ENAME                JOB                      DEPTNO
---------- -------------------- -------------------- ----------
DNAME                LOC
-------------------- --------------------
       103 TURNER               SALESMAN                     30
SALES                CHENNAI

       105 BLAKE                MANAGER                      30
SALES                CHENNAI

       107 ALLEN                SALESMAN                     30
SALES                CHENNAI


     EMPNO ENAME                JOB                      DEPTNO
---------- -------------------- -------------------- ----------
DNAME                LOC
-------------------- --------------------
       108 WARD                 SALESMAN                     30
SALES                CHENNAI

       109 MARTIN               SALESMAN                     30
SALES                CHENNAI

       110 JAMES                CLERK                        30
SALES                CHENNAI

15 rows selected.
```

12) Display the ENAME, DNAME of employees. Use RIGHT OUTER JOIN.

```
SELECT ENAME, DNAME
FROM EMP
RIGHT OUTER JOIN DEPT
ON EMP.DEPTNO = DEPT.DEPTNO;
```

Output:-

```
SQL> SELECT ENAME, DNAME
  2  FROM EMP
  3  RIGHT OUTER JOIN DEPT
  4  ON EMP.DEPTNO = DEPT.DEPTNO;

ENAME                DNAME
-------------------- --------------------
SCOTT                IT
TURNER               SALES
SMITH                IT
BLAKE                SALES
JONES                IT
ALLEN                SALES
WARD                 SALES
MARTIN               SALES
JAMES                SALES
MILLER               HR
CLARK                HR

ENAME                DNAME
-------------------- --------------------
ADAMS                IT
SARAH                HR
RAMESH               IT
KING                 HR

15 rows selected.

SQL> |
```

13) Display the EMPNO, DNAME, LOC of employees. Use FULL OUTER JOIN.

```
SELECT EMPNO, DNAME, LOC
FROM EMP
FULL OUTER JOIN DEPT
ON EMP.DEPTNO = DEPT.DEPTNO;
```

Output:-

```
SQL> SELECT EMPNO, DNAME, LOC
  2  FROM EMP
  3  FULL OUTER JOIN DEPT
  4  ON EMP.DEPTNO = DEPT.DEPTNO;

     EMPNO DNAME                LOC
---------- -------------------- --------------------
       102 IT                   BANGALORE
       103 SALES                CHENNAI
       104 IT                   BANGALORE
       105 SALES                CHENNAI
       106 IT                   BANGALORE
       107 SALES                CHENNAI
       108 SALES                CHENNAI
       109 SALES                CHENNAI
       110 SALES                CHENNAI
       111 HR                   HYDERABAD
       112 HR                   HYDERABAD

     EMPNO DNAME                LOC
---------- -------------------- --------------------
       113 IT                   BANGALORE
       114 HR                   HYDERABAD
       115 IT                   BANGALORE
       101 HR                   HYDERABAD

15 rows selected.

SQL> |
```